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%let path=F:\empiricalfinance\LSV1994;

libname value "&path.\data";
libname result "&path.\result";
libname stasum "&path.\statistic summary";

filename odsout "&path\result";
ods html path=odsout body="LSV_1994_2011.htm" (title="LSV from 1994 to 2011");

data monrel;
set value.monthly;
where stkcd like '60%' or stkcd like '001%' or stkcd like '000%'; /*选出
所有上海和深圳主板A股*/
year=input(substr(left(Trdmnt),1,4),4.); /*提取年度*/
month=input(substr(left(Trdmnt),6,2),2.); /*提取月度*/
rename Msmvttl=marketvalue Mretwd=return Mclsprc=clprice;
keep stkcd year month Msmvttl Mretwd Mclsprc;
run;
data monre2; set monrel;
if year>=1994; /*从1994年开始*/
if return^=.; /*数据集value.monthly中，提取出无缺省值的信息*/
marketvalue = marketvalue * 1000; /*国泰安数据库的总市值数据单位：千元*/
run;
proc sort data=monre2; by stkcd year month; run;
/*数据集monre2，包含符合条件个股的月度收益率、月末收盘价和月末总股本数据*/

/*填充有实际值首尾月份之间的缺省月度的资本收益率（首为其第一次收益率有值日；尾统一为
2013年9月）*/
data monre_everym1;
set monre2(keep=stkcd year month);
by stkcd year month;
if first.stkcd;
run; /*提取每只股票第一个交易月度*/
data monre_everym2; set monre_everym1;
endyear=2013;
endmonth=10;
run; /*统一最后一个月度为2013年10月*/
data monre_everym3;
merge monre_everym1 monre_everym2;
by stkcd;
run; /*数据集monre_everym3，包含每只股票的第一个交易月度和最后一个交易月度*/
data monre_everym4; set monre_everym3;
monthgap = (endyear-year-1)*12 + (12-month) + endmonth + 1; /*从开始到结束
的总月数计算*/
drop endyear endmonth;
run; /*数据集monre_everym4，包含每只股票的第一个交易月度和持续交易月度数（以日历计
算）*/
data monre_everym5; set monre_everym4;
do i=1 to monthgap;
output;
if month=12 then do; month=1; year=year+1; end;
else month=month+1;

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end;
drop monthgap i;
run; /*数据集monre_everym5, 包含每只股票的持续交易月度（以日历计算）*/
data monre_everym6;
merge monre2 monre_everym5;
by stkcd year month;
run; /*数据集monre_everym6, 包含完整月度的资本收益率（停牌月度为缺省值）*/

/*得到构造组合时停牌和特别处理的股票，以及收盘价低于1元的股票。在对价值投资指标排序前
将这三个股票集合剔除。*/
data value.nontrade;
set monre_everym6;
where month=6 and return=.;
keep stkcd year;
run; /*数据集value.nontrade, 生成构造组合时停牌各股记录*/

data value.pricelessone;
set monre_everym6;
where month=6 and clprice between 0 and 1;
if return^=.;
keep year stkcd;
run; /*数据集value.pricelessone, 生成构造组合时股价小于1（非停牌）各股记录*/

/*计算价值投资的指标：BM*/
data finance1; set value.LSV; keep stkcd year equity; run;
data finance2; set finance1; where equity^=.; run;
/*数据集finance2, 包含财务指标的原始值数据*/
data proxy1;
set monre2(drop=return);
where month=6; /*选择6月月末的数据（即用构造组合时的总市值计算各个价值投资指标）*/
year=year-1; /*将时间调整至与财务数据一致*/
drop month;
run;
data proxy2;
merge finance2(in=id) proxy1;
by stkcd year;
if id;
bm = equity / marketvalue; /*计算价值投资指标*/
keep stkcd year bm;
run; /*数据集proxy2, 包含计算的价值投资指标最原始数据（未进行任何剔除处理）*/
/*然后，进行剔除处理。注意：由于构造组合时用的是上一年数据，年度值需要做减1调整。*/
data nontrade; set value.nontrade;
year=year-1; /*将年度调整到和财务数据一致*/
run;
data special; set value.special;
year=year-1; /*将年度调整到和财务数据一致*/
run;
data pricelessone; set value.pricelessone;
year=year-1; /*将年度调整到和财务数据一致*/
run;
proc sort data=nontrade; by stkcd year; run;

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proc sort data=special; by stkcd year; run;
proc sort data=pricelessone; by stkcd year; run;
proc sort data=proxy2; by stkcd year; run;
data proxy3;
merge proxy2(in=id1) nontrade(in=id2) special(in=id3)
pricelessone(in=id4);
by stkcd year;
if id1=1 and id2=0 and id3=0 and id4=0;
run; /*数据集proxy3, 包含剔除了购买时点停牌、特别处理、收盘价小于1的股票的个股各年
价值投资指标*/
data proxy4; set proxy3;
where bm>0;
run; /*数据集proxy4, 仅提取正的价值投资指标*/
proc sort data=proxy3 out=bm_all; by year stkcd; run;
proc sort data=proxy4 out=bm; by year stkcd; run;

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/*生成排序宏。并对价值投资指标进行排序。*/
%macro rank_single(infile=,valueproxy=,outfile=);
proc sort data=&infile out=x1; by year stkcd; run; /*分组前进行排序*/
proc rank data=x1 out=x2 groups=10; /*每年进行一次对价值投资指标的分组*/
var &valueproxy;
by year;
ranks r_&valueproxy;
run;
data x3; set x2;
r_&valueproxy = r_&valueproxy + 1; /*分成10组, 组序为1-10*/
label r_&valueproxy = "&valueproxy.的升序标识";
drop &valueproxy;
run;
proc sort data=x3 out=&outfile; by stkcd year; run;
/*数据集r_x, 包含个股每年排序的组序数据, 并以代码、年排序*/

proc datasets lib=work nolist; delete x1-x3; run; quit;
%mend;
%rank_single(infile=bm,valueproxy=bm,outfile=r_bm);

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/*生成每年提取的具体股票代码、简称的宏*/
%macro stock(infile=,valueproxy=,outfile=);
data stock1; set &infile; run;
data stock2;
merge value.stkcd_name stock1(in=id); /*提取股票简称*/
by stkcd;
if id;
run;
proc sort data=stock2 out=&outfile; by year r_&valueproxy; run;
%mend;
%stock(infile=r_bm,valueproxy=bm,outfile=result.stock_bm);

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%macro stock_allrank(infile=,valueproxy=,outfile=);
data stock1; set &infile;

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run;
data stock2; set stock1;
format group_&valueproxy $6.; informat group_&valueproxy $6.;
if r_&valueproxy=1 then group_&valueproxy='成长股'; /*对组别命名*/
if r_&valueproxy=10 then group_&valueproxy='价值股';
run;
proc sort data=stock2 out=&outfile; by year r_&valueproxy; run;
%mend;
%stock_allrank(infile=r_bm,valueproxy=bm,outfile=result.stock_bm_allrank); /*这里有全部rank的股票，为后面table5计算所有rank做准备*/

/*生成计算所有股票第i年收益的宏*/
%macro period(beginyear=,endyear=2013,time=); /*time即为第i年年数：
1,2,3,4,5...*/
data acreturn1; set monre_everym6;
by stkcd year month;
retain acreturn;
if return=. then return=0; /*将停牌月度的回报数据设为0*/
if first.stkcd then acreturn=1+return;
else acreturn=acreturn*(1+return); /*计算累计收益*/
drop return clprice;
run; /*数据集acreturn1，包含各股从起始月度开始计算的累计收益数据*/
data acreturn2; set acreturn1;
year=year+1;
rename acreturn=acreturn1;
drop marketvalue;
run; /*数据集acreturn2，为了计算第i年的年收益*/
data acreturn3; set acreturn1;
year=year+&time;
rename marketvalue=marketvalue1;
drop acreturn;
run; /*数据集acreturn3，为了得到构造组合时的总市值（以便加权计算）*/
proc sort data=acreturn1; by stkcd year month; run;
proc sort data=acreturn2; by stkcd year month; run;
proc sort data=acreturn3; by stkcd year month; run;
data acreturn4;
merge acreturn1(in=id) acreturn2 acreturn3;
by stkcd year month;
if id;
run;
data acreturn5; set acreturn4;
valuereturn = acreturn/acreturn1 - 1; /*计算第i年的年收益*/
run; /*数据集acreturn5，包含有计算得到的第i年年收益和构造组合时的总市值数据*/
data acreturn6; set acreturn5;
if month=6; /*提取6月份的数值*/
if year>=&beginyear+&time+1; /*则第i年的年收益数据从(beginyear+i+1)年开始*/
label marketvalue1=' ';
drop month acreturn acreturn1 marketvalue;
run; /*数据集acreturn6，构造组合第i年后的6月末时，确定的第i年年收益，以及构造组合时的总市值（此时的year表示第i年后的年数）*/

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data return_&time.year; set areturn6;
year=year-&time-1; /*将year调整到与财务数据一致*/
run;
/*
数据集return_iyear, 包含所有股票在每年6月末开始的第i年的收益值, 和该年的总市值; year
和财务指标一致,
即该年的前一年。
*/
%mend;
%period(beginyear=1994,time=1);
%period(beginyear=1994,time=2);
%period(beginyear=1994,time=3);
%period(beginyear=1994,time=4);
%period(beginyear=1994,time=5);

/*****
*****
*
*
***** table1
*****
******/
/*构造一个输出宏*/

/*生成结果测试的宏*/
/*根据分组标识和未来第i年的收益值进行测试*/
%macro
test(endyear=2013,infile=,valueproxy=,outfile=,tablenum=,print=1);
%do time=1 %to 5;
data test1; set &infile;
if year <= &endyear-&time-1; /* (endyear-i-1) 年的价值投资指标分组才有持有第i
年的数据*/
run;
proc sort data=test1; by stkcd year; run; /*对数据集test1处理, 为merge做准
备*/
data test2;
merge test1(in=id) return_&time.year;
by stkcd year;
if id;
run;
proc sort data=test2; by year r_&valueproxy; run;
/*数据集test2, 包含每年每个分组的各股第i年收益率, 按年、分组标识排序*/

proc means data=test2 noprint; /*市值加权*/
var valuereturn;
weight marketvalue1;
by year r_&valueproxy;
output out=test3 mean=mwreturn; /*每年计算一次加权平均*/
run;
proc sort data=test3; by r_&valueproxy year; run;
proc means data=test3 noprint;
var mwreturn;
by r_&valueproxy;
output out=test4 mean=R&time; /*再对所有年份平均*/

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run;
data return_&time; set test4; drop _type_ _freq_; run;
/*数据集return_x, 包含加权平均下, 每年组合在第i年收益值的平均*/
%end;
data result_&valueproxy; /*加权收益结果*/
merge return_1 return_2 return_3 return_4 return_5;
by r_&valueproxy;
AR = (R1+R2+R3+R4+R5)/5; /*算术平均*/
CR = (1+R1)*(1+R2)*(1+R3)*(1+R4)*(1+R5)-1; /*几何平均*/
run;
proc transpose data=result_&valueproxy out=result.&outfile;
id r_&valueproxy;
run;
* print;
title "Table &tablenum (1994 - 2011)";
title2 "Sort variable: &valueproxy";
%if &print=1 %then %do;
proc print;
format _numeric_ comma10.4;
run; %end;
%mend;
/*table1, panelA*/
%test(valueproxy=bm,infile=r_bm,outfile=result_bm,tablenum=1);

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%macro figure_onedim(infile=,title=,title1=,vbar=,sumvar=);
/*绘图的全局变量*/
Goptions reset=global gunit=pct cback=white border
htitle=6 htext=3 ftext='宋体';
title c=black h=4 "&title";
title2 c=black h=4 "&title1";
proc gchart data=&infile;
vbar &vbar / discrete type=sum sumvar=&sumvar;
run;
quit;
%mend;
%figure_onedim(infile=result_bm,title=figure1A,title1=returns in year 1
after formation,vbar=r_bm,sumvar=R1);
%figure_onedim(infile=result_bm,title=figure1A,title1=returns in year 2
after formation,vbar=r_bm,sumvar=R2);
%figure_onedim(infile=result_bm,title=figure1A,title1=returns in year 3
after formation,vbar=r_bm,sumvar=R3);
%figure_onedim(infile=result_bm,title=figure1A,title1=returns in year 4
after formation,vbar=r_bm,sumvar=R4);
%figure_onedim(infile=result_bm,title=figure1A,title1=returns in year 5
after formation,vbar=r_bm,sumvar=R5);
%figure_onedim(infile=result_bm,title=figure1A,title1=average
returns,vbar=r_bm,sumvar=AR);
%figure_onedim(infile=result_bm,title=figure1A,title1=compounded
returns,vbar=r_bm,sumvar=CR);

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/*计算价值投资的指标: CP*/
data financel; set value.LSV; keep stkcd year NCFope; run;

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data finance2; set financel; where NCFope^=.; run;
/*数据集finance2, 包含财务指标的原始值数据*/
data proxy1;
set monre2(drop=return);
where month=6; /*选择6月月末的数据（即用构造组合时的总市值计算各个价值投资指标）*/
year=year-1; /*将时间调整至与财务数据一致*/
drop month;
run;
data proxy2;
merge finance2(in=id) proxy1;
by stkcd year;
if id;
cp = NCFope / marketvalue; /*计算价值投资指标*/
keep stkcd year cp;
run; /*数据集proxy2, 包含计算的价值投资指标最原始数据（未进行任何剔除处理）*/
/*然后, 进行剔除处理。注意: 由于构造组合时用的是上一年数据, 年度值需要做减1调整。*/
data nontrade; set value.nontrade;
year=year-1; /*将年度调整到和财务数据一致*/
run;
data special; set value.special;
year=year-1; /*将年度调整到和财务数据一致*/
run;
data pricelessone; set value.pricelessone;
year=year-1; /*将年度调整到和财务数据一致*/
run;
proc sort data=nontrade; by stkcd year; run;
proc sort data=special; by stkcd year; run;
proc sort data=pricelessone; by stkcd year; run;
proc sort data=proxy2; by stkcd year; run;
data proxy3;
merge proxy2(in=id1) nontrade(in=id2) special(in=id3)
pricelessone(in=id4);
by stkcd year;
if id1=1 and id2=0 and id3=0 and id4=0;
run; /*数据集proxy3, 包含剔除了购买时点停牌、特别处理、收盘价小于1的股票的个股各年
价值投资指标*/
data proxy4; set proxy3;
where cp>0;
run; /*数据集proxy4, 仅提取正的价值投资指标*/
proc sort data=proxy3 out=cp_all; by year stkcd; run;
proc sort data=proxy4 out=cp; by year stkcd; run;

%rank_single(infile=cp,valueproxy=cp,outfile=r_cp);
%stock(infile=r_cp,valueproxy=cp,outfile=result.stock_cp);
/*table1, panelB*/
%test(valueproxy=cp,infile=r_cp,outfile=result_cp,tablenum=1);

%figure_onedim(infile=result_cp,title=figure1B,title1=returns in year 1
after formation,vbar=r_cp,sumvar=R1);
%figure_onedim(infile=result_cp,title=figure1B,title1=returns in year 2
after formation,vbar=r_cp,sumvar=R2);
%figure_onedim(infile=result_cp,title=figure1B,title1=returns in year 3
after formation,vbar=r_cp,sumvar=R3);
%figure_onedim(infile=result_cp,title=figure1B,title1=returns in year 4
after formation,vbar=r_cp,sumvar=R4);

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%figure_onedim(infile=result_cp,title=figure1B,title1=returns in year 5
after formation,vbar=r_cp,sumvar=R5);
%figure_onedim(infile=result_cp,title=figure1B,title1=average
returns,vbar=r_cp,sumvar=AR);
%figure_onedim(infile=result_cp,title=figure1B,title1=compounded
returns,vbar=r_cp,sumvar=CR);

/*计算价值投资的指标: EP*/
data finance1; set value.LSV; keep stkcd year netprofit; run;
data finance2; set finance1; where netprofit^=.; run;
/*数据集finance2, 包含财务指标的原始值数据*/
data proxy1;
set monre2(drop=return);
where month=6; /*选择6月月末的数据（即用构造组合时的总市值计算各个价值投资指标）*/
year=year-1; /*将时间调整至与财务数据一致*/
drop month;
run;
data proxy2;
merge finance2(in=id) proxy1;
by stkcd year;
if id;
ep = netprofit / marketvalue; /*计算价值投资指标*/
keep stkcd year ep;
run; /*数据集proxy2, 包含计算的价值投资指标最原始数据（未进行任何剔除处理）*/
/*然后, 进行剔除处理。注意: 由于构造组合时用的是上一年数据, 年度值需要做减1调整。*/
data nontrade; set value.nontrade;
year=year-1; /*将年度调整到和财务数据一致*/
run;
data special; set value.special;
year=year-1; /*将年度调整到和财务数据一致*/
run;
data pricelessone; set value.pricelessone;
year=year-1; /*将年度调整到和财务数据一致*/
run;
proc sort data=nontrade; by stkcd year; run;
proc sort data=special; by stkcd year; run;
proc sort data=pricelessone; by stkcd year; run;
proc sort data=proxy2; by stkcd year; run;
data proxy3;
merge proxy2(in=id1) nontrade(in=id2) special(in=id3)
pricelessone(in=id4);
by stkcd year;
if id1=1 and id2=0 and id3=0 and id4=0;
run; /*数据集proxy3, 包含剔除了购买时点停牌、特别处理、收盘价小于1的股票的个股各年
价值投资指标*/
data proxy4; set proxy3;
where ep>0;
run; /*数据集proxy4, 仅提取正的价值投资指标*/
proc sort data=proxy3 out=ep_all; by year stkcd; run;
proc sort data=proxy4 out=ep; by year stkcd; run;

%rank_single(infile=ep,valueproxy=ep,outfile=r_ep);
%stock(infile=r_ep,valueproxy=ep,outfile=result.stock_ep);

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/*table1, panelC*/
%test(valueproxy=ep,infile=r_ep,outfile=result_ep,tablenum=2);

*figure;
%figure_onedim(infile=result_ep,title=figure1C,title1=returns in year 1
after formation,vbar=r_ep,sumvar=R1);
%figure_onedim(infile=result_ep,title=figure1C,title1=returns in year 2
after formation,vbar=r_ep,sumvar=R2);
%figure_onedim(infile=result_ep,title=figure1C,title1=returns in year 3
after formation,vbar=r_ep,sumvar=R3);
%figure_onedim(infile=result_ep,title=figure1C,title1=returns in year 4
after formation,vbar=r_ep,sumvar=R4);
%figure_onedim(infile=result_ep,title=figure1C,title1=returns in year 5
after formation,vbar=r_ep,sumvar=R5);
%figure_onedim(infile=result_ep,title=figure1C,title1=average
returns,vbar=r_ep,sumvar=AR);
%figure_onedim(infile=result_ep,title=figure1C,title1=compounded
returns,vbar=r_ep,sumvar=CR);

/*****
*****
*
***** table2
*****
*****/
/*对两个指标同时进行排序并测试*/

/*ep_bm双变量测试*/
/*生成一个双排序宏*/
%macro
rank_double(infile1=infile2,valueproxy1=valueproxy2,outfile=);
proc sort data=&infile1 out=rank1; by year stkcd; run;
proc sort data=&infile2 out=rank2; by year stkcd; run;
data rank3;
merge rank1(in=id1) rank2(in=id2);
by year stkcd;
if id1 and id2; /*同时有值, 才能进行双排序*/
run;
/*按年生成分位数, 并得出分组标识。*/
proc univariate data=rank3 noprint;
by year;
var &valueproxy1;
output out=&valueproxy1._dec pctlpts = 30 70 pctlpre=&valueproxy1.dec;
run;
proc univariate data=rank3 noprint;
by year;
var &valueproxy2;
output out=&valueproxy2._dec pctlpts = 30 70 pctlpre=&valueproxy2.dec;
run;
data rank4; merge rank3 &valueproxy1._dec &valueproxy2._dec; by year; run;
data rank5; set rank4;
if &valueproxy1 <= &valueproxy1.dec30 then _1=1;
else if &valueproxy1 <= &valueproxy1.dec70 then _1=2;
else _1=3;

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if &valueproxy2 <= &valueproxy2.dec30 then _2=1;
    else if &valueproxy2 <= &valueproxy2.dec70 then _2=2;
    else _2=3;
r_&valueproxy1._&valueproxy2 = 10*_1 + _2;
drop _1 _2;
run;
data rank6; set rank5;
drop &valueproxy1.dec30 &valueproxy1.dec70
    &valueproxy2.dec30 &valueproxy2.dec70
    &valueproxy1 &valueproxy2;
run;
proc sort data=rank6 out=&outfile; by r_&valueproxy1._&valueproxy2; run;
%mend;
/*对ep_bm双排序*/
%rank_double(infile1=ep,infile2=bm,valueproxy1=ep,valueproxy2=bm,outfile=r_ep_bm);

/*生成每年提取价值组和成长组的具体股票代码、简称的宏（针对双排序情况）*/
%macro stock_double(infile=,doublevalueproxy=,outfile=);
data stock1; set &infile;
where r_&doublevalueproxy=11 or r_&doublevalueproxy=33; /*提取数据集中组序
为11和33的部分*/
run;
data stock2; set stock1;
format group_&doublevalueproxy $6.; informat group_&doublevalueproxy $6.;
if r_&doublevalueproxy=11 then group_&doublevalueproxy='成长股'; /*对组别
命名*/
if r_&doublevalueproxy=33 then group_&doublevalueproxy='价值股';
run;
proc sort data=stock2 out=&outfile; by year r_&doublevalueproxy; run;
%mend;
%stock_double(infile=r_ep_bm,doublevalueproxy=ep_bm,outfile=result.stock_ep_bm);
/*对ep_bm双变量进行测试*/
/*table2, panelD*/
%test(valueproxy=ep_bm,infile=r_ep_bm,outfile=result_ep_bm,tablenum=2);

* figure 2D;
%macro figure_twodim(infile=,var1=,var2=,title=,title2=,z=,zmax=);
data figure1; set &infile;
r_&var1 = floor(r_&var1._&var2./10); label r_&var1 = "&var1.的升序标识";
r_&var2 = r_&var1._&var2 - 10 * r_&var1; label r_&var2 = "&var2.的升序标识";
drop r_&var1._&var2;
run;

/*绘图的全局变量*/
goptions device=actximg;
goptions noborder;
goptions gunit=pct htitle=6 ftitle="albany amt/bold" htext=4.25
ftext="albany amt/bold";

title ls=1.5 h=4 c=black f="黑体" "&title";
title2 h=4 c=black f="黑体" "&title2";

```

```

pattern1 v=solid color=cx9999ff; /* 浅蓝 */
pattern2 v=solid color=cx993366; /* 紫色 */
pattern3 v=solid color=cxffffcc; /* 明黄 */

/*绘图*/
proc gchart data=figure1;
block r_&var1 / discrete type=sum sumvar=&z shape=block nolegend
  blockmax=&zmax
  group=r_&var2
  subgroup=r_&var2 /* 这个控制着色 */
  cframe=white caxis=black coutline=black;
run;

quit;
%mend;
%figure_twodim(infile=result_ep_bm,var1=ep,var2=bm,title=figure2D,
  title2=returns in year 1 after formation,z=R1,zmax=0.3);
%figure_twodim(infile=result_ep_bm,var1=ep,var2=bm,title=figure2D,
  title2=returns in year 2 after formation,z=R2,zmax=0.3);
%figure_twodim(infile=result_ep_bm,var1=ep,var2=bm,title=figure2D,
  title2=returns in year 3 after formation,z=R3,zmax=0.3);
%figure_twodim(infile=result_ep_bm,var1=ep,var2=bm,title=figure2D,
  title2=returns in year 4 after formation,z=R4,zmax=0.3);
%figure_twodim(infile=result_ep_bm,var1=ep,var2=bm,title=figure2D,
  title2=returns in year 5 after formation,z=R5,zmax=0.3);
%figure_twodim(infile=result_ep_bm,var1=ep,var2=bm,title=figure2D,
  title2=average return,z=AR,zmax=0.3);
%figure_twodim(infile=result_ep_bm,var1=ep,var2=bm,title=figure2D,
  title2=compounded return,z=CR,zmax=2);

/*bm_cp双变量测试*/
/*对bm_cp双排序*/
%rank_double(infile1=bm,infile2=cp,valueproxy1=bm,valueproxy2=cp,outfile=r_bm_cp);
/*对bm_cp双变量进行测试*/
/*table2, panelE*/
%test(valueproxy=bm_cp,infile=r_bm_cp,outfile=result_bm_cp,tablenum=2);
%figure_twodim(infile=result_bm_cp,var1=bm,var2=cp,title=figure2E,
  title2=returns in year 1 after formation,z=R1,zmax=0.3);
%figure_twodim(infile=result_bm_cp,var1=bm,var2=cp,title=figure2E,
  title2=returns in year 2 after formation,z=R2,zmax=0.3);
%figure_twodim(infile=result_bm_cp,var1=bm,var2=cp,title=figure2E,
  title2=returns in year 3 after formation,z=R3,zmax=0.3);
%figure_twodim(infile=result_bm_cp,var1=bm,var2=cp,title=figure2E,
  title2=returns in year 4 after formation,z=R4,zmax=0.3);
%figure_twodim(infile=result_bm_cp,var1=bm,var2=cp,title=figure2E,
  title2=returns in year 5 after formation,z=R5,zmax=0.3);
%figure_twodim(infile=result_bm_cp,var1=bm,var2=cp,title=figure2E,
  title2=average return,z=AR,zmax=0.3);
%figure_twodim(infile=result_bm_cp,var1=bm,var2=cp,title=figure2E,
  title2=compounded return,z=CR,zmax=2);

```

```

/*****
*****
*
*
***** table3
*****
*****/
/*只对每个构造组合时点时，总市值前50%的股票进行处理*/
data size1;
set monre_everym6;
where month=6 and return^=.;
drop month return clprice;
run;
proc sort data=size1 out=size2; by year stkcd; run;
proc univariate data=size2 noprint;
by year;
var marketvalue;
output out=size3 pctlpts=50 pctlpre=dec; /*50%分位数*/
run;
data size4;
merge size2 size3;
by year;
run;
data value.size; set size4;
where marketvalue >= dec50; /*选出每年总市值大于或等于50%的股票*/
keep year stkcd;
run; /*数据集value.size, 包含每年总市值大于或等于50%的股票。以下市这个数据集对bm、
cp、ep进行进一步筛选。*/

data size_bm1; /*BM*/
merge bm(in=id1) value.size(in=id2);
by year stkcd;
if id1 and id2; /*选择两个数据集的交叉，即为之前的bm数据集再加上条件：总市值位于市
场前50%*/
run;

data size_cp1; /*CP*/
merge cp(in=id1) value.size(in=id2);
by year stkcd;
if id1 and id2; /*选择两个数据集的交叉，即为之前的cp数据集再加上条件：总市值位于市
场前50%*/
run;

data size_ep1; /*EP*/
merge ep(in=id1) value.size(in=id2);
by year stkcd;
if id1 and id2; /*选择两个数据集的交叉，即为之前的ep数据集再加上条件：总市值位于市
场前50%*/
run;

/*ep_bm双变量测试*/
%rank_double(infile1=size_ep1,infile2=size_bm1,valueproxy1=ep,valueprox
y2=bm,outfile=r_size_ep_bm);
/*table3, panelD*/
%test(valueproxy=ep_bm,infile=r_size_ep_bm,outfile=result_size_ep_bm,ta

```

```

blenum=3);
/*bm_cp双变量测试*/
%rank_double(infile=size_bm1,infile2=size_cp1,valueproxy1=bm,valueprox
y2=cp,outfile=r_size_bm_cp);
/*table3, panelE*/
%test(valueproxy=bm_cp,infile=r_size_bm_cp,outfile=result_size_bm_cp,ta
blenum=3);

/*****
*****
Table 3.1 取size50%的样本, run table1 的panel A
*****
*****/
%rank_single(infile=size_bm1,valueproxy=bm,outfile=size_r_bm);
%test(valueproxy=bm,infile=size_r_bm,outfile=result_size_bm,tablenum=3.
1);

/*****
*****
*
*
*****
*****/
/*对各股第1年的回报进行回归分析, 解释变量为B/M, SIZE, CP(+), DCP, EP(+), DEP*/
data reg_ret1; set return_1year; run;
data finastkcd; set value.finastkcd; run; /*为了将金融行业公司剔除*/
data ifaccount; set value.LSV;
keep stkcd year;
run; /*为了将会计年度无财报的观测剔除*/
data nontrade; set value.nontrade;
year = year-1; /*将年度调整到和财务数据一致*/
run;
data special; set value.special;
year = year-1; /*将年度调整到和财务数据一致*/
run;
data pricelessone; set value.pricelessone;
year = year-1; /*将年度调整到和财务数据一致*/
run;
proc sort data=finastkcd; by stkcd year; run;
proc sort data=ifaccount; by stkcd year; run;
proc sort data=nontrade; by stkcd year; run;
proc sort data=special; by stkcd year; run;
proc sort data=pricelessone; by stkcd year; run;
proc sort data=reg_ret1; by stkcd year; run;
data reg_ret2;
merge reg_ret1(in=id1) finastkcd(in=id2) ifaccount(in=id3) nontrade(in=id4)
special(in=id5) pricelessone(in=id6);
by stkcd year;
if id1=1 and id2=0 and id3=1 and id4=0 and id5=0 and id6=0;
run;
/*数据集reg_ret2, 包含剔除了金融企业、会计年无财报(会计年下一年新上市)、购买时点停
牌、特别处理、收盘价小于1的股票的个股第1年回报*/
data size; set reg_ret2;
size = (marketvalue1/1000000); /*SIZE (以百万人民币表示)*/
drop marketvalue1 valuereturn;

```

```

run;
data reg_ret3; set reg_ret2;
size = log(marketvalue1/1000000); /*回归用SIZE*/
drop marketvalue1;
run;
proc sort data=reg_ret3; by year stkcd; run;

data reg_bm1; set bm; run; /*BM*/
data reg_cp1; set cp_all;
if cp<0 and cp^=. then cp=0;
run; /*CP(+)*
data reg_dcp1; set cp_all;
if cp<0 and cp^=. then dcp=1;
if cp>0 and cp^=. then dcp=0;
drop cp;
run; /*DCP*/
data reg_ep1; set ep_all;
if ep<0 and ep^=. then ep=0;
run; /*EP(+)*
data reg_dep1; set ep_all;
if ep<0 and ep^=. then dep=1;
if ep>0 and ep^=. then dep=0;
drop ep;
run; /*DEP*/

data reg1;
merge reg_ret3 reg_bm1 reg_cp1 reg_dcp1 reg_ep1 reg_dep1;
by year stkcd;
if year<=2011; /*会计年2011的组合构造年为2012，完整观测一年的截止期为2013年6月，
此为最后一个会计年度*/
run;

/*构造一个回归宏*/
%macro regression(var1=,var2=,var3=,var4=,outfile=);
proc reg data=reg1 noprint outest=result1;
model valuereturn = &var1 &var2 &var3 &var4;
by year;
run;
data result2; set result1;
keep year intercept &var1 &var2 &var3 &var4;
run;
proc means data=result2 noprint;
var intercept &var1 &var2 &var3 &var4;
output out=result_mean1(drop=_freq_ _type_) mean=;
run;
data result_mean2;
informat item $6.; format item $6.;
set result_mean1;
item = '平均值';
run;
proc means data=result2 noprint;
var intercept &var1 &var2 &var3 &var4;
output out=result_t1(drop=_freq_ _type_) t=;
run;
data result_t2;

```

```

informat item $6.; format item $6.;
set result_t1;
item = 't值';
run;
data &outfile; set result_mean2 result_t2; run;
%mend;

%regression(var1=bm,outfile=reg_ret_bm);
%regression(var1=size,outfile=reg_ret_size);
%regression(var1=cp,outfile=reg_ret_cp);
%regression(var1=ep,outfile=reg_ret_ep);
%regression(var1=bm,var2=cp,var3=dcg,outfile=reg_ret_bm_cp_dcp);
%regression(var1=bm,var2=size,var3=cp,var4=dcg,outfile=reg_ret_bm_size_cp_dcp);
%regression(var1=bm,var2=size,var3=ep,var4=dep,outfile=reg_ret_bm_size_ep_dep);

data result.reg_result;
set reg_ret_bm reg_ret_size reg_ret_cp reg_ret_bm_cp_dcp
    reg_ret_bm_size_cp_dcp reg_ret_ep reg_ret_bm_size_ep_dep;
run;
* table4 print;
    title "Table 4";
    title2 "Regression of Returns on Characteristics for All Firms";
proc print data=result.reg_result;
    format intercept bm size cp dcg ep dep comma10.3;
run;

/*****
*****
*
*
*
*****
*****
*/
table5
/对价值股和成长股进行基础变量、过去表现和实际未来表现的检测*/
/*基础变量*/
/*计算价值投资的指标: SP*/
data finance1; set value.LSV; keep stkcd year revenue; run;
data finance2; set finance1; where revenue^=.; run;
/*数据集finance2, 包含财务指标的原始值数据*/
data proxy1;
set monre2(drop=return);
where month=6; /*选择6月月末的数据（即用构造组合时的总市值计算各个价值投资指标）*/
year=year-1; /*将时间调整至与财务数据一致*/
drop month;
run;
data proxy2;
merge finance2(in=id) proxy1;
by stkcd year;
if id;
sp = revenue / marketvalue; /*计算价值投资指标*/
keep stkcd year sp;
run; /*数据集proxy2, 包含计算的价值投资指标最原始数据（未进行任何剔除处理）*/
/*然后, 进行剔除处理。注意: 由于构造组合时用的是上一年数据, 年度值需要做减1调整。*/

```

```

data nontrade; set value.nontrade;
year=year-1; /*将年度调整到和财务数据一致*/
run;
data special; set value.special;
year=year-1; /*将年度调整到和财务数据一致*/
run;
data pricelessone; set value.pricelessone;
year=year-1; /*将年度调整到和财务数据一致*/
run;
proc sort data=nontrade; by stkcd year; run;
proc sort data=special; by stkcd year; run;
proc sort data=pricelessone; by stkcd year; run;
proc sort data=proxy2; by stkcd year; run;
data proxy3;
merge proxy2(in=id1) nontrade(in=id2) special(in=id3)
pricelessone(in=id4);
by stkcd year;
if id1=1 and id2=0 and id3=0 and id4=0;
run; /*数据集proxy3, 包含剔除了购买时点停牌、特别处理、收盘价小于1的股票的个股各年
价值投资指标*/
data proxy4; set proxy3;
where sp>0;
run; /*数据集proxy4, 仅提取正的价值投资指标*/
proc sort data=proxy4 out=sp; by year stkcd; run;

/*计算价值投资的指标: DP*/
data financel; set value.LSV; keep stkcd year dividend; run;
data finance2; set financel; where dividend^=.; run;
/*数据集finance2, 包含财务指标的原始值数据*/
data proxy1;
set monre2(drop=return);
where month=6; /*选择6月月末的数据（即用构造组合时的总市值计算各个价值投资指标）*/
year=year-1; /*将时间调整至与财务数据一致*/
drop month;
run;
data proxy2;
merge finance2(in=id) proxy1;
by stkcd year;
if id;
dp = dividend / marketvalue; /*计算价值投资指标*/
keep stkcd year dp;
run; /*数据集proxy2, 包含计算的价值投资指标最原始数据（未进行任何剔除处理）*/
/*然后, 进行剔除处理。注意: 由于构造组合时用的是上一年数据, 年度值需要做减1调整。*/
data nontrade; set value.nontrade;
year=year-1; /*将年度调整到和财务数据一致*/
run;
data special; set value.special;
year=year-1; /*将年度调整到和财务数据一致*/
run;
data pricelessone; set value.pricelessone;
year=year-1; /*将年度调整到和财务数据一致*/
run;
proc sort data=nontrade; by stkcd year; run;
proc sort data=special; by stkcd year; run;

```



```

proc sort data=pricelessone; by stkcd year; run;
proc sort data=proxy2; by stkcd year; run;
data proxy3;
merge proxy2(in=id1) nontrade(in=id2) special(in=id3)
pricelessone(in=id4);
by stkcd year;
if id1=1 and id2=0 and id3=0 and id4=0;
run; /*数据集proxy3, 包含剔除了购买时点停牌、特别处理、收盘价小于1的股票的个股各年
价值投资指标*/
data proxy4; set proxy3;
where dp>0;
run; /*数据集proxy4, 仅提取正的价值投资指标*/
proc sort data=proxy4 out=dp; by year stkcd; run;

/*计算基础指标: 形成期6月的成交额与换手率*/
data trade1; set value.monthly;
where stkcd like '60%' or stkcd like '001%' or stkcd like '000%'; /*选出
所有上海和深圳主板A股*/
year=input(substr(left(Trdmnt),1,4),4.); /*提取年度*/
month=input(substr(left(Trdmnt),6,2),2.); /*提取月度*/
rename Mnvaltrd=tradesize Msmvosd=amarketvalue;
label Mnvaltrd=' ' Msmvosd=' ';
keep stkcd year month Mnvaltrd Msmvosd;
run;
data trade2; set trade1;
if year>=1994; /*从1994年开始*/
amarketvalue = amarketvalue * 1000; /*国泰安数据库的流通市值数据单位: 千元*/
run;
proc sort data=trade2; by stkcd year month; run;
data trade3; set trade2;
where month=6;
year=year-1; /*将年度调整到基础财务数据的年度, 即会计年度*/
turnover = 100 * tradesize/amarketvalue;
drop month;
run;
proc sort data=trade3 out=trade; by year stkcd; run;

/*分价值组和成长组计算基础指标*/
proc sort data=result.stock_bm out=stock_bm; by year stkcd; run;
proc sort data=ep; by year stkcd; run;
proc sort data=cp; by year stkcd; run;
proc sort data=sp; by year stkcd; run;
proc sort data=dp; by year stkcd; run;
proc sort data=bm; by year stkcd; run;
proc sort data=size; by year stkcd; run;
proc sort data=trade; by year stkcd; run;
data fundamental1;
merge stock_bm(in=id) ep cp sp dp bm size trade;
by year stkcd;
if id;
run;
proc sort data=fundamental1 out=fundamental2; by year r_bm stkcd; run;

```

```

/*生成一个计算股票分组跨年平均的宏*/
%macro aveacrossyear(infile=,var=,outfile=);
proc means data=&infile noprint;
var &var;
by year r_bm;
output out=fundamental3(drop=_type_ _freq_) mean=&var;
run;
proc sort data=fundamental3; by r_bm year; run;
proc means data=fundamental3 noprint;
var &var;
by r_bm;
output out=&outfile(drop=_type_ _freq_) mean=&var;
run;
%mend;
%aveacrossyear(infile=fundamental2,var=ep,outfile=fundamental_ep);
%aveacrossyear(infile=fundamental2,var=cp,outfile=fundamental_cp);
%aveacrossyear(infile=fundamental2,var=sp,outfile=fundamental_sp);
%aveacrossyear(infile=fundamental2,var=dp,outfile=fundamental_dp);
%aveacrossyear(infile=fundamental2,var=bm,outfile=fundamental_bm);
%aveacrossyear(infile=fundamental2,var=size,outfile=fundamental_size);
%aveacrossyear(infile=fundamental2,var=tradesize,outfile=fundamental_tradesize);
%aveacrossyear(infile=fundamental2,var=turnover,outfile=fundamental_turnover);

data fundamental_result;
set fundamental_ep; set fundamental_cp; set fundamental_sp; set fundamental_dp;
set fundamental_bm; set fundamental_size; set fundamental_tradesize; set
fundamental_turnover;
run;
proc transpose data=fundamental_result out=result.fundamental;
id r_bm;
run;
* table5 panelA print;
title "Table 5.1 all rank";
title2 "Fundamental Variables";
proc print data=result.fundamental;
format _numeric_ comma20.3;
run;

```

/*分价值组和成长组计算过去三年的净利润、经营性现金流、营业收入的年化增长，以及累计收益*/

/*净利润和营业收入增长数据从1997年到2012年*/

/*经营性现金流增长数据从2001年到2012年*/

/*累计收益数据从1997年到2012年*/

/*以上年份均是指会计年度，特别地是，累计收益数据实际是从1998年6月开始有值。*/

/*生成一个计算相关指标增长的宏*/

```
%macro gross(var=,outfile=,relative1=,relative2=);
```

/*relative1、relative2表示与当年的相对年数差，且满足：relative1<relative2<=0，或者relative2>relative1>=0*/

/*例如，relative1=-3，relative2=0，表示过去3年的增长；relative=1，relative=3表示未来的第1年到第3年，2年的增长*/

```

%let interval = %eval(&relative2-&relative1); /*时间间隔*/
%if &relative2 > 0 %then %do;
    %let num1 = &relative1;
    %let num2 = &relative2;
%end;
%if &relative1 < 0 %then %do;
    %let num1 = %eval(-(&relative1));
    %let num2 = %eval(-(&relative2));
%end;
data gross1; set value.LSV;
keep stkcd year &var;
if &var > 0; /*正的财务指标才有讨论增长的意义*/
run;
data gross2; set gross1;
year=year+&interval;
rename &var=&var.1;
run;
data gross3;
merge gross1(in=id) gross2;
by stkcd year;
run;
data gross4; set gross3;
&var._&num1._&num2 = (&var/&var.1)**(1/&interval)-1; /*几何平均值，即年化值*/
drop &var &var.1;
run;
/*调整年度数，使之表示当年时点*/
%if &relative2 > 0 %then %do;
    data gross5; set gross4;
    year = year - &relative2; /*如果relative2大于0，将年度调整到起始点*/
run;
%end;
%if &relative1 < 0 %then %do;
    data gross5; set gross4; run; /*如果relative1小于0，不调整起始点*/
%end;
data &outfile; set gross5; where &var._&num1._&num2 ^= .; run;
%mend;
%gross(var=netprofit,outfile=gross_netprofit_3_0,relative1=-3,relative2=0);
%gross(var=revenue,outfile=gross_revenue_3_0,relative1=-3,relative2=0);
%gross(var=NCFope,outfile=gross_NCFope_3_0,relative1=-3,relative2=0);

/*计算构造组合时，各股过去三年的累计收益*/
data grossreturn1; set monre_everym6; by stkcd year month; retain acreturn;
if return=. then return=0;
if first.stkcd then acreturn=1+return;
else acreturn=acreturn*(1+return);
keep stkcd year month acreturn;
run;
data grossreturn2; set grossreturn1;
where month=6;
year=year-1; /*将时间调整到与财务数据一致*/
drop month;
run;
data grossreturn3; set grossreturn2;

```

```

year=year+3;
rename acreturn=acreturn1;
run;
data grossreturn4;
merge grossreturn2(in=id) grossreturn3;
by stkcd year;
if id;
run;
data grossreturn5; set grossreturn4;
where year>=1997; /*为了和净利润、营业收入一致，从1997年开始*/
return_3_0 = (acreturn/acreturn1) - 1;
drop acreturn acreturn1;
run;
data gross_return_3_0; set grossreturn5; where return_3_0^=.; run;

proc sort data=result.stock_bm out=stock_bm; by year stkcd; run;
proc sort data=gross_netprofit_3_0; by year stkcd; run;
proc sort data=gross_revenue_3_0; by year stkcd; run;
proc sort data=gross_NCFope_3_0; by year stkcd; run;
proc sort data=gross_return_3_0; by year stkcd; run;
data gross_post;
merge stock_bm(in=id) gross_netprofit_3_0 gross_revenue_3_0
gross_NCFope_3_0 gross_return_3_0;
by year stkcd;
if id;
run;
proc sort data=gross_post; by year r_bm stkcd; run;

/*求过去表现的跨年平均*/
%aveacrossyear(infile=gross_post,var=netprofit_3_0,outfile=gross_p_3_0)
;
%aveacrossyear(infile=gross_post,var=revenue_3_0,outfile=gross_r_3_0);
%aveacrossyear(infile=gross_post,var=NCFope_3_0,outfile=gross_cf_3_0);
%aveacrossyear(infile=gross_post,var=return_3_0,outfile=gross_ret_3_0);

data gross_post_result;
set gross_p_3_0; set gross_r_3_0; set gross_cf_3_0; set gross_ret_3_0;
run;
proc transpose data=gross_post_result out=result.gross_post;
id r_bm;
run;
/*table5, panelB*/
* table5 panelB print;
  title "Table 5";
  title2 "Past Performance-Growth Rates and Past Returns";
proc print data=result.gross_post;
  format _numeric_ comma10.4;
run;

/*分价值组和成长组计算未来三年的净利润、经营性现金流、营业收入的年化增长，以及未来一年
后的两年净利润、经营性现金流、营业收入的年化增长*/
/*净利润和营业收入增长数据从1994年到2009年*/
/*经营性现金流增长数据从1998年到2009年*/
/*以上年份均是指会计年度*/

```

```

%gross(var=netprofit,outfile=gross_netprofit_0_3,relative1=0,relative2=
3);
%gross(var=revenue,outfile=gross_revenue_0_3,relative1=0,relative2=3);
%gross(var=NCFope,outfile=gross_NCFope_0_3,relative1=0,relative2=3);
%gross(var=netprofit,outfile=gross_netprofit_1_3,relative1=1,relative2=
3);
%gross(var=revenue,outfile=gross_revenue_1_3,relative1=1,relative2=3);
%gross(var=NCFope,outfile=gross_NCFope_1_3,relative1=1,relative2=3);

proc sort data=result.stock_bm out=stock_bm; by year stkcd; run;
proc sort data=gross_netprofit_0_3; by year stkcd; run;
proc sort data=gross_revenue_0_3; by year stkcd; run;
proc sort data=gross_NCFope_0_3; by year stkcd; run;
proc sort data=gross_netprofit_1_3; by year stkcd; run;
proc sort data=gross_revenue_1_3; by year stkcd; run;
proc sort data=gross_NCFope_1_3; by year stkcd; run;
data gross_future;
merge stock_bm(in=id) gross_netprofit_0_3 gross_revenue_0_3
gross_NCFope_0_3
gross_netprofit_1_3 gross_revenue_1_3
gross_NCFope_1_3;
by year stkcd;
if id;
run;
proc sort data=gross_future; by year r_bm stkcd; run;

/*求过去表现的跨年平均*/
%aveacrossyear(infile=gross_future,var=netprofit_0_3,outfile=gross_p_0_
3);
%aveacrossyear(infile=gross_future,var=revenue_0_3,outfile=gross_r_0_3)
;
%aveacrossyear(infile=gross_future,var=NCFope_0_3,outfile=gross_cf_0_3)
;
%aveacrossyear(infile=gross_future,var=netprofit_1_3,outfile=gross_p_1_
3);
%aveacrossyear(infile=gross_future,var=revenue_1_3,outfile=gross_r_1_3)
;
%aveacrossyear(infile=gross_future,var=NCFope_1_3,outfile=gross_cf_1_3)
;

data gross_future_result;
set gross_p_0_3; set gross_r_0_3; set gross_cf_0_3;
set gross_p_1_3; set gross_r_1_3; set gross_cf_1_3;
run;
proc transpose data=gross_future_result out=result.gross_future;
id r_bm;
run;
/*table5, panelC*/
* table5 panelC print;
title "Table 5";
title2 "Future Performance";
proc print data=result.gross_future;
format _numeric_ comma10.4;
run;

```

```

/*****
table 6, 计算1年, 2年和3年的buy and hold 收益率率
*****/

/*生成计算所有股票第i年buy and hold 收益的宏*/
%macro buyhold(beginyear=,endyear=2013,time=); /*time即为第i年年数:
1,2,3,4,5...*/

data acretun1; set monre_everym6;
by stkcd year month;
retain acretun;
if return=. then return=0; /*将停牌月度的回报数据设为0*/
if first.stkcd then acretun=1+return;
else acretun=acretun*(1+return); /*计算累计收益*/
drop return clprice;
run; /*数据集acretun1, 包含各股从起始月度开始计算的累计收益数据*/

/*data acretun2; set acretun1;
year=year+&time;
rename acretun=acretun1;
drop marketvalue;
run; /*数据集acretun2, 为了计算第i年的年收益*/

data acretun3; set acretun1;
year=year+&time;
rename marketvalue=marketvalue1 acretun=acretun1;
run; /*数据集acretun3, 为了得到构造组合时的总市值（以便加权计算）和组后的buy and
hold 收益率*/

proc sort data=acretun1; by stkcd year month; run;
proc sort data=acretun3; by stkcd year month; run;

data acretun4;
merge acretun1(in=id) acretun3;
by stkcd year month;
if id;
run;

data acretun5; set acretun4;
valuereturn = acretun/acretun1 - 1; /*计算buy and hold 到第i年的年收益*/
run; /*数据集acretun5, 包含有计算得到的第i年年收益和构造组合时的总市值数据*/

data acretun6; set acretun5;
if month=6; /*提取6月份的数值*/
if year>=&beginyear+&time+1; /*则第i年的年收益数据从(beginyear+i+1)年开始*/
label marketvalue1=' ';
drop month acretun acretun1 marketvalue;
run; /*数据集acretun6, 构造组合第i年后的6月末时, 确定的第i年年收益, 以及构造组合
时的总市值（此时的year表示第i年后的年数）*/
data buyhold_&time.year; set acretun6;

```

```

year=year-&time-1; /*将year调整到与财务数据一致*/
run;
/*
数据集return_iyear, 包含所有股票在每年6月末开始的第i年的收益值, 和该年的总市值; year
和财务指标一致,
即该年的前一年。
*/
%mend;
%buyhold(beginyear=1994,time=1);
%buyhold(beginyear=1994,time=2);
%buyhold(beginyear=1994,time=3);

%macro table6(endyear=2013,infile=,valueproxy=);
%do time=1 %to 3;
data test1; set &infile;
if year <= &endyear-&time-1; /* (endyear-i-1) 年的价值投资指标分组才有持有第i
年的数据*/
run;
proc sort data=test1; by stkcd year; run; /*对数据集test1处理, 为merge做准
备*/
data test2;
merge test1(in=id) buyhold_&time.year;
by stkcd year;
if id;
run;
proc sort data=test2; by year r_&valueproxy; run;
/*数据集test2, 包含每年每个分组的各股第i年收益率, 按年、分组标识排序*/

proc means data=test2 noprint; /*市值加权*/
var valuereturn;
weight marketvalue1;
by year r_&valueproxy;
output out=test3 mean=mwreturn&time; /*每年计算一次加权平均*/
run;

data buyhold_&time; set test3; drop _type_ _freq_; run;
/*数据集buyhold_x, 包含加权平均下, 每年组合buy and hold i年收益值的平均*/

proc transpose data=buyhold_&time out=buyhold1_&time; by year; id
r_&valueproxy; run;

data buyhold2_&time; set buyhold1_&time;
year=year+1;
run;
proc sort data=buyhold2_&time; by year _name_;
run;
%mend;

data result.buyhold_&valueproxy;
set buyhold2_1 buyhold2_2 buyhold2_3;
run;
%mend;

/*table6, pane3*/

```

```

%table6(valueproxy=bm,infile=r_bm);
%table6(valueproxy=cp,infile=r_cp);

data year_1_bm; set result.buyhold_bm; if _NAME_="mwreturn1"; run;
data year_1_bm1; set year_1_bm; bm12=mean(_1,_2); bm910=mean(_9,_10);
minus101=_10-_1; minus910_21=bm910-bm12; run;

data year_1_cp; set result.buyhold_cp; if _NAME_="mwreturn1"; run;
data year_1_cp1; set year_1_cp; cp12=mean(_1,_2); cp910=mean(_9,_10);
minus101=_10-_1; minus910_21=cp910-cp12; run;

* table6 print;
  title "Table 6";
  title2 "Buy&hold_BM";
  proc print data=result.buyhold_bm;
    format year 10. _1 _2 _3 _4 _5 _6 _7 _8 _9 _10 comma10.4 ;
  run;
* table6 print;
  title "Table 6";
  title2 "Buy&hold_CP";
  proc print data=result.buyhold_cp;
    format year 10. _1 _2 _3 _4 _5 _6 _7 _8 _9 _10 comma10.4 ;
  run;

%figure_onedim(infile=year_1_bm1,title=figure6A(sorted by bm),title1=port
1 returns in year 1 after formation,vbar=year,sumvar=_1);
%figure_onedim(infile=year_1_bm1,title=figure6B(sorted by bm),title1=
port 10 returns in year 1 after formation,vbar=year,sumvar=_10);
%figure_onedim(infile=year_1_bm1,title=figure6C(sorted by bm),title1=port
12 returns in year 1 after formation,vbar=year,sumvar=bm12);
%figure_onedim(infile=year_1_bm1,title=figure6D(sorted by bm),title1=port
910 returns in year 1 after formation,vbar=year,sumvar=bm910);
%figure_onedim(infile=year_1_bm1,title=figure6E(sorted by bm),title1=port
10 minus port 1 returns in year 1 after formation,vbar=year,sumvar=minus101);
%figure_onedim(infile=year_1_bm1,title=figure6F(sorted by bm),title1=port
910 minus port 12 returns in year 1 after
formation,vbar=year,sumvar=minus910_21);

%figure_onedim(infile=year_1_cp1,title=figure6A(sorted by cp),title1=port
1 returns in year 1 after formation,vbar=year,sumvar=_1);
%figure_onedim(infile=year_1_cp1,title=figure6B(sorted by cp),title1=
port 10 returns in year 1 after formation,vbar=year,sumvar=_10);
%figure_onedim(infile=year_1_cp1,title=figure6C(sorted by cp),title1=port
12 returns in year 1 after formation,vbar=year,sumvar=cp12);
%figure_onedim(infile=year_1_cp1,title=figure6D(sorted by cp),title1=port
910 returns in year 1 after formation,vbar=year,sumvar=cp910);
%figure_onedim(infile=year_1_cp1,title=figure6E(sorted by cp),title1=port
10 minus port 1 returns in year 1 after formation,vbar=year,sumvar=minus101);
%figure_onedim(infile=year_1_cp1,title=figure6F(sorted by cp),title1=port
910 minus port 12 returns in year 1 after
formation,vbar=year,sumvar=minus910_21);

```



```

/*****
*****
*
*
*****
*****
table7
*****
*****/
/*计算不同市况下的组合表现*/
/*以下程序段的时间是交易实际时间，会将财务数据的年度进行调整*/
/*首先，构造一个按上月市值加权的指数，并将之作为基准。若某一个月停牌，则该月不计入指数；
若上一个月停牌而本月有交易，该月也不计入指数*/
data index1; set monre_everym6; keep stkcd year month return marketvalue;
run;
data index2; set index1;
t = (year-1994)*12 + month; /*时间变量，为了后面能够将前一个月的总市值对应上*/
run;
data index3; set index2;
t = t+1;
rename marketvalue=marketvalue1;
label marketvalue=' ';
drop year month return;
run;
data index4;
merge index2(in=id) index3;
by stkcd t;
run;
data index5; set index4; drop t marketvalue;
where return^=. and marketvalue1^=.; /*满足条件：若某一个月停牌，则该月不计入
指数；若上一个月停牌而本月有交易，该月也不计入指数*/
if t >= 18; /*从1995年6月开始计算指数*/
run;
proc sort data=index5; by year month stkcd; run;
/*开始计算指数*/
proc means data=index5 noprint;
var return;
by year month;
weight marketvalue1;
output out=value.vwindex(drop=_type_ _freq_) mean=vwindex;
run;
data value.vwindex; set value.vwindex; label vwindex=' '; run;

/*然后，取指数表现最好的前25个月，表现最差的后25个月*/
data month1; set value.vwindex; where vwindex>=0; if year=1995 and month=6
then delete; run;
proc sort data=month1; by descending vwindex; run;
data best; set month1;
if _n_<=25;
run; /*指数表现最好的25个月*/
data positive; set month1;
if _n_>25;
run; /*指数为正，但剔除了表现最好25个月的月份*/
data month2; set value.vwindex; where vwindex<0; if year=1995 and month=6
then delete; run;
proc sort data=month2; by vwindex; run;
data worst; set month2;
if _n_<=25;
run; /*指数表现最差的25个月*/

```

```

data negative; set month2;
if _n_>25;
run; /*指数为负，但剔除了表现最差25个月的月份*/

/*将按BM以月分组出来的股票与前述四个月度分类进行合并*/
data month_rbm1; set monre2;
year=year-1; /*将时间调整到与财务数据一致*/
run;
/*找到构造组合时的总市值*/
data month_rbm2;
merge r_bm(in=id) month_rbm1;
by stkcd year;
if id;
run;
data month_rbm3; set month_rbm2;
where month=6;
keep stkcd year r_bm marketvalue;
rename marketvalue=marketvalue1;
label marketvalue=' ';
run;
/*每只股票的组别和实际对应的年月*/
data month_rbm4(drop=i); set month_rbm3;
year=year+1; /*组合构造于下一年度*/
month=7; /*组合表现从7月份开始*/
do i=1 to 12;
    output;
    if month=12 then do; year=year+1; month=1; end;
    else month=month+1;
end;
run;

/*各种月度分类下，各个分组的平均收益*/
%macro month_class(month_class=,itemname=,outfile=);
/*月度分组和分组标识合并*/
proc sort data=&month_class; by year month; run;
proc sort data=month_rbm4; by year month; run;
data month_class1;
merge &month_class(in=id drop=vwindex) month_rbm4;
by year month;
if id;
run;
/*再和该月收益值合并*/
proc sort data=month_class1; by year month stkcd; run;
proc sort data=monre2 out=month_class2; by year month stkcd; run;
data month_class3;
merge month_class1(in=id) month_class2(keep=stkcd year month return);
by year month stkcd;
if id;
label return=' ';
run;
/*每月每组内的各股收益加权平均*/
proc sort data=month_class3; by year month r_bm; run;
proc means data=month_class3 noprint;
var return;

```

```

weight marketvalue1;
by year month r_bm;
output out=month_class4(drop=_type_ _freq_) mean=return;
run;
/*跨月度平均以及相关计算*/
/*跨月度平均*/
proc sort data=month_class4; by r_bm year month; run;
proc means data=month_class4 noprint;
var return;
by r_bm;
output out=month_class5(drop=_type_ _freq_) mean=return;
run;
data month_class6;
informat item $15.; format item $15.;
set month_class5;
item = r_bm;
drop r_bm;
run;
/*指数跨月度平均*/
proc means data=&month_class noprint;
var vwindex;
output out=month_index1(drop=_type_ _freq_) mean=return;
run;
data month_index2;
informat item $15.; format item $15.;
set month_index1;
item = 'index';
run;
/*9,10组与1,2组收益之差, 及其t统计量*/
proc sort data=month_class4; by year month r_bm; run;
proc transpose data=month_class4 out=month_class7(drop=_name_);
id r_bm;
by year month;
run;
data month_class8; set month_class7;
return = mean(_9, _10)-mean(_1, _2);
keep year month _1 _2 _9 _10 return;
run;
proc means data=month_class8 noprint;
var return;
output out=month_class9(drop=_type_ _freq_) mean=value_glamour
t=t_statistic;
run;
/*结果合并*/
data month_class10; set month_class6 month_index2; run;
proc transpose data=month_class10 out=month_class11;
id item;
run;
data month_class12; set month_class11; _name_=&itemname; run;
proc sql;
create table &outfile as select
month_class12.*,month_class9.*
from month_class12,month_class9;
quit;
%mend;
%month_class(month_class=best,itemname='B25',outfile=month_result_best)

```

```

;
%month_class(month_class=positive,itemname='P91',outfile=month_result_posi
tive);
%month_class(month_class=negative,itemname='N79',outfile=month_result_ne
gative);
%month_class(month_class=worst,itemname='W25',outfile=month_result_wors
t);

data result.month_class;
set month_result_worst month_result_negative month_result_positive
month_result_best;
run;
/*table7, panel1B*/
* table6 print;
title "Table 7";
title2 "Performance of Portfolios in Best and Worst Times";
proc print data=result.month_class;
format _numeric_ comma10.4;
run;

/*****
*****
***** table8
*****
*****/
/*利用传统的风险测量模型检验组合风险*/
/*市场风险: beta*/
/*首先, 计算组合形成后一年的指数累计收益*/
data beta_index1; set value.vwindex; retain acreturn;
if _n_=1 then acreturn=1; /*从1995年6月底开始计算指数的累计收益*/
else acreturn=acreturn*(1+vwindex);
drop vwindex;
run;
data beta_index2; set beta_index1; where month=6; drop month; run;
data beta_index3; set beta_index2;
year=year+1;
rename acreturn=acreturn1;
run;
data beta_index4;
merge beta_index2(in=id) beta_index3;
by year;
if id;
run;
data beta_index5; set beta_index4;
return_index = acreturn/acreturn1 -1;
if return_index^=.;
drop acreturn acreturn1;
run;
data beta_index6; set beta_index5;
year=year-2; /*将时间调整至与财务数据一致*/
run;
/*然后, 计算各个组合每年的形成一年后的累计收益*/
%macro return_year(infile=,var=,outfile=);
data beta_port1; set &infile;

```

```

if year <= 2011; /*2011年的BM指标分组才有形成后一年的数据*/
run;
proc sort data=beta_port1; by stkcd year; run; /*对数据集beta_port1处理,
为merge做准备*/
data beta_port2;
merge beta_port1(in=id) return_1year;
by stkcd year;
if id;
run;
proc sort data=beta_port2; by year &var; run; /*数据集beta_port2, 包含每年
每个分组的各股第1年收益率, 按年、分组标识排序*/
proc means data=beta_port2 noprint; /*市值加权*/
var valuereturn;
weight marketvalue1;
by year &var; /*每年计算一次加权平均*/
output out=beta_port3(drop=_type_ _freq_) mean=portreturn;
run;
proc transpose data=beta_port3 out=beta_port4(drop=_name_);
id &var;
by year;
run;
/*合并组合分年收益和对应年度指数收益*/
data &outfile;
merge beta_port4 beta_index6;
by year;
run;
%mend;
%return_year(infile=r_bm,var=r_bm,outfile=beta_reg_bm);
%return_year(infile=r_cp,var=r_cp,outfile=beta_reg_cp);
%return_year(infile=r_ep,var=r_ep,outfile=beta_reg_ep);

/*各组收益对指数收益进行回归*/
%macro beta_regression(infile=,outfile=);
%do time=1 %to 10;
proc reg data=&infile noprint outest=beta_result&time(keep=return_index
rename=(return_index=_&time));
model _&time = return_index;
run;
quit;
%end;
data &outfile;
informat item $15.; format item $15.;
set beta_result1; set beta_result2; set beta_result3; set beta_result4; set
beta_result5;
set beta_result6; set beta_result7; set beta_result8; set beta_result9; set
beta_result10;
item = 'beta';
run;
%mend;
%beta_regression(infile=beta_reg_bm,outfile=beta_result_bm);
%beta_regression(infile=beta_reg_cp,outfile=beta_result_cp);
%beta_regression(infile=beta_reg_ep,outfile=beta_result_ep);

/*求各组收益的标准差*/
%macro beta_std(infile=,outfile=);

```

```

%do time=1 %to 10;
proc means data=&infile noprint;
var _&time;
output out=beta_std&time(drop=_type_ _freq_) std=_&time;
run;
%end;
data &outfile;
informat item $15.; format item $15.;
set beta_std1; set beta_std2; set beta_std3; set beta_std4; set beta_std5;
set beta_std6; set beta_std7; set beta_std8; set beta_std9; set beta_std10;
item = 'Std';
run;
%mend;
%beta_std(infile=beta_reg_bm,outfile=beta_std_bm);
%beta_std(infile=beta_reg_cp,outfile=beta_std_cp);
%beta_std(infile=beta_reg_ep,outfile=beta_std_ep);

/*合并结果*/
data result.risk_cp;
set beta_result_cp beta_std_cp;
run;
/*table8, panel1*/
* table8 panel1 print;
  title "Table 8";
  title2 "Traditional Risk Measures for Portfolios (Sorted by cp)";
data result.risk_bm;
set beta_result_bm beta_std_bm;
run;
/*table8, panel3*/
* table8 panel3 print;
  title "Table 8";
  title2 "Traditional Risk Measures for Portfolios (Sorted by bm)";
proc print data=result.risk_bm;
  format _numeric_ comma10.4;
run;
data result.risk_ep;
set beta_result_ep beta_std_ep;
run;
/*table8, panel4*/
* table8 panel4 print;
  title "Table 8";
  title2 "Traditional Risk Measures for Portfolios (Sorted by ep)";
proc print data=result.risk_ep;
  format _numeric_ comma10.4;
run;

ods html close;
ods html;

/*****选择股票提取出来*****/
ods html path=odsout body="Selected Stock by BM Group.htm" (title="Selected
Stock by BM Group");
title "Year-by-Year Selected Stock by BM Group";
title2 "总市值数据来自当年6月底的市场数据，账面值数据来自前一年底的财报；在当年6月底
排序并对股票分组";

```

```

proc sort data=value.special_name out=special_name; by year stkcd; run;
proc sort data=result.stock_bm out=stock_bm; by year stkcd; run;
data stock_bm; set stock_bm;
year=year+1; /*将年份调整到构造组合的时点*/
run;
data selected1;
merge stock_bm(in=id) special_name;
by year stkcd;
if id;
run;
data selected2; set selected1;
if stknme_all^=' ' then stknme = stknme_all;
drop stknme_all;
run;
proc sort data=selected2; by year r_bm stkcd; run;
proc print data=selected2 noobs label;
label year='排序的年份';
run;
ods html close;
ods html;

```